

1 Solution — GIAM 8.2

Problem 2

1.1 Problem

Prove that

$$f(x) = c + \frac{(x-a)(d-c)}{b-a}$$

provides a bijection from the interval $[a, b]$ to the interval $[c, d]$.

1.2 Setup — Assumptions and Goal

Throughout, assume $a < b$ and $c < d$, so that $[a, b]$ and $[c, d]$ are non-degenerate closed intervals and $b - a > 0$, $d - c > 0$. (If $a = b$ the formula is undefined; if $c = d$ the formula collapses to a constant and is not injective.)

To show $f : [a, b] \rightarrow [c, d]$ is a bijection, we verify three things:

1. f maps $[a, b]$ into $[c, d]$ (codomain check).
2. f is *injective*.
3. f is *surjective* onto $[c, d]$.

The function f is affine: $f(x) = mx + k$ where the *slope* is

$$m = \frac{d-c}{b-a} > 0,$$

and the *intercept-style constant* is $k = c - ma$. Affine maps with $m \neq 0$ are well-known to be bijections of $\mathbb{R} \rightarrow \mathbb{R}$; the work below is just verifying that, restricted to $[a, b]$, the image is exactly $[c, d]$.

1.3 Endpoint Check

Direct evaluation:

$$f(a) = c + \frac{(a-a)(d-c)}{b-a} = c + 0 = c,$$

$$f(b) = c + \frac{(b-a)(d-c)}{b-a} = c + (d-c) = d.$$

So f sends the left endpoint a to the left endpoint c , and the right endpoint b to the right endpoint d . This already tells us the image will fill out $[c, d]$; the next two steps make it precise.

1.4 Codomain — f Maps $[a, b]$ into $[c, d]$

For $x \in [a, b]$, we have $0 \leq x - a \leq b - a$. Multiplying through by the positive number $(d - c)/(b - a)$:

$$0 \leq \frac{(x-a)(d-c)}{b-a} \leq \frac{(b-a)(d-c)}{b-a} = d - c.$$

Adding c to all three sides,

$$c \leq c + \frac{(x-a)(d-c)}{b-a} \leq d, \quad \text{i.e.,} \quad c \leq f(x) \leq d.$$

So $f(x) \in [c, d]$.

1.5 Injectivity

Suppose $f(x_1) = f(x_2)$. Subtracting c from both sides and clearing the common positive factor $(d - c)/(b - a)$:

$$\frac{(x_1 - a)(d - c)}{b - a} = \frac{(x_2 - a)(d - c)}{b - a} \implies x_1 - a = x_2 - a \implies x_1 = x_2.$$

The cancellation is legal because $d - c \neq 0$ and $b - a \neq 0$. So f is injective.

1.6 Surjectivity — Constructing the Preimage

Let $y \in [c, d]$. We need $x \in [a, b]$ with $f(x) = y$. Solving $f(x) = y$ algebraically for x :

$$y = c + \frac{(x - a)(d - c)}{b - a}$$

$$y - c = \frac{(x - a)(d - c)}{b - a}$$

$$\frac{(y - c)(b - a)}{d - c} = x - a$$

$$x = a + \frac{(y - c)(b - a)}{d - c}.$$

This x exists as a real number because $d - c \neq 0$. We must check that $x \in [a, b]$.

For $y \in [c, d]$, we have $0 \leq y - c \leq d - c$. Multiplying by the positive number $(b - a)/(d - c)$:

$$0 \leq \frac{(y - c)(b - a)}{d - c} \leq b - a.$$

Adding a : $a \leq x \leq b$. So $x \in [a, b]$.

Finally, verify $f(x) = y$:

$$f(x) = c + \frac{(x - a)(d - c)}{b - a} = c + \frac{1}{b - a} \cdot \frac{(y - c)(b - a)}{d - c} \cdot (d - c) = c + (y - c) = y. \quad \checkmark$$

So f is surjective onto $[c, d]$.

1.7 Conclusion

$f : [a, b] \rightarrow [c, d]$ is well-defined, injective, and surjective — hence a bijection. The *inverse* bijection is

$$f^{-1} : [c, d] \rightarrow [a, b], \quad f^{-1}(y) = a + \frac{(y - c)(b - a)}{d - c},$$

which has the same shape as f with the roles of (a, b) and (c, d) swapped — as expected, since “linear interpolation from one interval to another” is its own inverse with roles swapped. ■

1.8 Remark — Geometric View

The graph of $y = f(x)$ is the unique straight line through the points (a, c) and (b, d) :

$$\frac{y - c}{x - a} = \frac{d - c}{b - a} \iff y = c + \frac{(x - a)(d - c)}{b - a}.$$

The slope $(d - c)/(b - a)$ is the *ratio of the interval lengths*. Restricting this line to $x \in [a, b]$ traces out the line segment from (a, c) to (b, d) , and projecting back to the y -axis covers $[c, d]$ exactly once. The bijection is just “linear interpolation”.

1.9 Remark — f as a Composition of Three Standard Bijections

The formula decomposes naturally as

$$[a, b] \xrightarrow{x \mapsto x - a} [0, b - a] \xrightarrow{u \mapsto u \cdot (d - c)/(b - a)} [0, d - c] \xrightarrow{v \mapsto v + c} [c, d].$$

Each arrow is an affine bijection (shift by $-a$, scale by the positive factor $(d - c)/(b - a)$, shift by $+c$). The composition is the formula for f , and bijectivity follows from the fact that each step is a bijection between the indicated intervals (and the composition of bijections is a bijection — Section 8.1 Problem 2).

This three-step view is the *reason* the formula has the shape it does: shift so the left endpoint is 0 , rescale so the right endpoint is $d - c$, shift so the left endpoint is c . The slope $(d - c)/(b - a)$ is precisely the rescaling factor that aligns the two interval lengths.

1.10 Remark — The Same Formula Bijects All Four Interval Types

Because f is *strictly increasing* and continuous, it maps each subinterval of $[a, b]$ onto the correspondingly-typed subinterval of $[c, d]$:

Domain restricted to	Image
$[a, b]$	$[c, d]$

(a, b)	(c, d)
$[a, b)$	$[c, d)$
$(a, b]$	$(c, d]$

Table 1.1

So the exact same formula provides a bijection $(a, b) \rightarrow (c, d)$ and either half-open analog. The endpoint inclusions are preserved by the bijection precisely because f sends endpoints to endpoints and is strictly monotone.

1.11 Remark — Why This Matters for Chapter 8

Section 8.2 is in the business of showing that “looking different” sets can have the *same* cardinality. Problem 2 is the workhorse for *all* finite real intervals:

Any two non-degenerate intervals of $\mathbb{R}$ are equinumerous.

Why? By the formula above, $[a, b] \equiv [c, d]$ for any choice of $a < b$ and $c < d$. Combined with the *projection-from-a-point* bijection in Section 8.2 (which biject the open interval $(-1, 1)$ with the entire real line $\mathbb{R}$, via the half-circle), one concludes

$$[a, b] \equiv [c, d] \equiv (a, b) \equiv \mathbb{R}.$$

So a centimeter-long interval has the same cardinality as a kilometer-long one — and either has the same cardinality as the whole real line. The formula in this problem is the first half of that surprising chain.

1.12 Remark — What If $c > d$? Or $c = d$?

The formula still makes algebraic sense as long as $b \neq a$, but:

- If $c > d$: the “interval $[c, d]$ ” is empty under the usual convention $[c, d] = \{x : c \leq x \leq d\}$. Writing $[d, c]$ instead, the formula gives a *decreasing* affine bijection $[a, b] \rightarrow [d, c]$; the proof above goes through with all inequality directions reversed.

- If $c = d$: the slope $(d - c)/(b - a) = 0$, so f is the constant function $x \mapsto c$. This is *not* a bijection from $[a, b]$ (a non-degenerate interval) to $\{c\}$.

The intended hypothesis is therefore $a < b$ and $c < d$, which the problem statement leaves implicit.